

Irregular Respiratory Motion Compensation for Liver Contrast-Enhanced Ultrasound via Transport-Based Motion Estimation

Peng Wan¹, Fang Chen¹, Wei Shao¹, Chunrui Liu, Yidan Zhang, Baojie Wen, Wentao Kong, and Daoqiang Zhang

Abstract—Contrast-enhanced ultrasound (CEUS) imaging has been widely applied for the detection and characterization of focal liver lesions (FLLs), for its ability to visualize the blood flow in real time. However, cyclic liver motion poses a great challenge to the recovery of perfusion curves as well as quantitative kinetic parameters estimation. Recently, a few gating methods have been proposed to eliminate unexpected intensity fluctuations by the breathing phase estimation, with the assumption that each breathing phase corresponds to a specific lesion position strictly. While practical liver motion tends to be irregular due to changes in the patient's underlying physiologic status, that is, the same phase might not correspond to the same position. To tackle the challenge of motion irregularity, we present a novel motion estimation-based respiratory compensation method, named RCME, which first estimates salient motion information through the framework of optimal transport (OT) by jointly modeling pixel intensity as well as their locations and then employs sparse subspace clustering (SSC) to identify the subset of frames acquired at the same position. Our proposed method is evaluated on 15 clinical CEUS sequences in both qualitative and quantitative manners. Experimental results demonstrate good performance on irregular liver motion compensation.

Index Terms—Liver contrast-enhanced ultrasound (CEUS), motion compensation, optimal transport (OT).

I. INTRODUCTION

CONTRAST-ENHANCED ultrasound (CEUS) has become a promising imaging modality for assessing the focal liver lesions (FLLs), from early screening [1] and differential diagnosis [2] to prognostic evaluation [3]. Compared with the conventional B-mode ultrasound (BUS) and other imaging techniques (e.g., contrast-enhanced MRI and contrast-enhanced CT), the biggest advantage of CEUS is to create an opportunity to observe the blood flow enhancement difference between lesions and surrounding healthy tissues (i.e., the parenchyma) in real time [4]. This is achieved through the injection of microbubble contrast agents (CAs), with a comparable size to red blood cells [5], which amplify the echoes from microvessels while suppressing the others. Generally, CEUS cine loop is the sequence of contiguous frames that record the complete perfusion process (i.e., the wash-in and wash-out of CAs). Clinically, diverse enhancement patterns associated with tumor angiogenesis have been used as major imaging biomarkers for the FLLs evaluation [6], [7]. Also, the most common quantitative description manner is to extract kinetic parameters from regional time-intensity curves (TICs), such as time to peak (TTP), peak intensity (PI), and area under curves (AUC) [8]. Accordingly, an accurate delineation of regional TICs is of particular importance.

However, accurate TICs acquisition might be challenging due to unexpected intensity fluctuations primarily caused by the breathing-induced liver motion [8], [9]. Previous respiratory compensation methods generally assume that lesion movement follows regular and cyclic motion patterns [10], [11], and there exists an associated breathing curve of which each phase corresponds to a specific lesion position accurately. Nonetheless, practical organ movements are usually impacted by a number of unpredictable factors [12], for example, the free breathing as well as cardiac movement. These factors are closely correlated with patients' physiological status (e.g., emotional changes, sudden cough, and muscle tension variety [13]), which are hard to control especially for the elderly or

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Peng Wan, Fang Chen, Wei Shao, and Daoqiang Zhang are with the MIIT Key Laboratory of Pattern Analysis and Machine Intelligence, College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing 211106, China (e-mail: dqzhang@nuaa.edu.cn).

Chunrui Liu, Yidan Zhang, Baojie Wen, and Wentao Kong are with the Department of Ultrasound, Drum Tower Hospital, The Affiliated Hospital of Nanjing University Medical School, Nanjing 210008, China (e-mail: breezewen@163.com).

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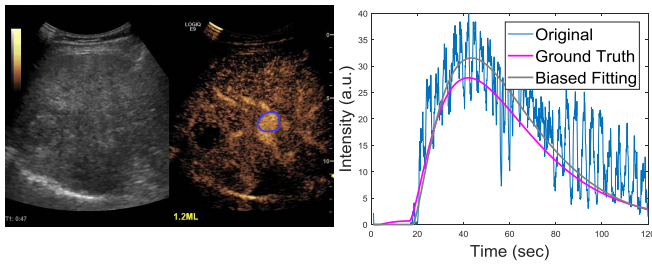


Fig. 1. Illustration of the biased fitting of regional TIC caused by irregular liver motion. Left: example of dual-mode CEUS frame, comprising the conventional B-mode tissue view and the simultaneous contrast-specific view. Right: original TIC with irregular intensity fluctuation (denoted by the blue line) and the biased kinetic curve recovery (denoted by the gray line) compared with the ground truth (denoted by the magenta line).

those who suffer from cardiovascular diseases [1]. Therefore, practical liver motion is characterized by irregularity, as shown in Fig. 1, and considerable irregular intensity fluctuations inevitably lead to a biased estimate of TICs kinetic parameters.

In order to alleviate the impact of lesion motion, considerable attempts have been investigated for compensation [14]–[18]. Among which, restricting patients to breathe shallowly even to hold breath during CEUS examination is one of the simplest approaches. Nonetheless, considering that the whole perfusion process usually takes at least 2–3 min [4], most patients are unlikely to keep fully sedated as required. To this end, various computer-aided compensation methods have been proposed to compensate for liver motion in 2-D CEUS sequences. Depending on whether to preserve all recorded contrast images, we can divide them into two categories: 1) image registration-based methods [19], [20] and 2) respiratory gating methods [10], [11].

Generally, registration-based methods attempt to align anatomical structures across the CEUS sequence. For the dual-view mode, estimation of 2-D translation (e.g., linear affine translation [21] or nonlinear deformation field [22], [23]) can be performed either on the contrast-specific mode or the simultaneous BUS mode. The former attempts to recover motion information directly from contrast sequences where the dramatical changes of pixel intensity are the mixture result of the CA injection and organ motion. Thus, point-based registration techniques (PBRTs) are more appropriate since local structure descriptors are less sensitive to changes in illumination, orientation, and contrast [12]. The latter is dedicated to estimating deformation fields based on grayscale images under the intensity constancy assumption. High-contrast anatomical structures play an important role in these methods, such as liver diaphragm and vessels. However, aligning each contrast frame to the template is time-consuming and computationally expensive. Actually, preserving all frames is not necessary for the recovery of kinetic curves due to the continuity of contrast concentration variation and high temporal resolution of the CEUS imaging.

As an alternative to registration, respiratory gating methods are designed to identify the subset of frames acquired at predefined breathing phases (e.g., the end-respiratory or the end-inspiratory phase). With the continuity assumption,

the final TIC is constructed by smoothing intensities of the preserved temporal points. Basically, gating methods depend on the accuracy of recovered 1-D respiratory signals over time. Expect for the online approach relying on external sensors attached to skin [24], [25], most offline gating techniques simplify this problem as extracting the low-dimensional motion component from BUS cine loop. Existing component analysis methods, including independent component analysis (ICA) [26], principal component analysis (PCA) [10], [27], and nonnegative matrix factorization (NMF) [28], have been exploited for respiratory kinetics curves estimation from successive BUS images. Based on the estimated respiratory curves, the end-of-inspiration and end-of-expiration frames are detected. However, respiration irregularity during the CEUS examination would impede the effectiveness of breathing phase detection, for the reason that the same phase does not necessarily correspond to the same lesion position caused by the varying breathing magnitudes [29]. Fayad *et al.* [25] proposed to simultaneously take magnitude and phase into account with a specified threshold that needs to be carefully adjusted for each subject.

In this article, we present a novel motion estimation-based respiratory compensation method for liver CEUS analysis, named RCME, specifically designed to address the challenge of motion irregularity. Specifically, we utilize the framework of optimal transport (OT) to explicitly model the continuous variation of tissue spatial distribution in B-mode sequences, local tissue expansions/contractions are captured by the mass-preserving mapping for tissues matching and displacement estimation [30] and then leveraging the multisubspace structure of the sequential motion matrix, and the CEUS subsequence corresponding to the reference point is identified by sparse subspace clustering (SSC) to restore TICs. Qualitatively, the effectiveness of our method is validated using the parametric perfusion imaging (PPI) formed with pixel-level TICs (detailed in Section IV). Quantitatively, we compare the compensation performance of our method with several competing respiration compensation techniques. The main innovations of our method can be summarized as follows.

- 1) We first introduce the transport-based method, OT, to estimate the motion information of dominant moving structures in the synchronous B-mode sequence. Combined with SSC, motion estimation naturally avoids the problem of inconsistency between respiratory phases and lesion positions in case of irregular liver motion.
- 2) The OT distance of tissue spatial distribution defines not only a metric space but also a Riemannian manifold that is consistent with the continuous liver motion. In addition, OT-derived transformation significantly enhances sample separability compared with the original pixel domain for its ability to jointly model pixel intensity as well as their locations.
- 3) Considering the strong structural similarity of neighboring frames, we choose the SSC to model the multisubspace structure of the sequence of estimated motion sequence, with the assumption that each motion matrix can only be represented by a sparse set of motion matrix from the same position.

The rest of this article is organized as follows. In Sections II and III, we introduce the studied data sets and our proposed RCME method, respectively. In Section IV, we give a detailed description of experimental setting and the implementation of competing methods. In Section V, the effectiveness of our proposed RCME is evaluated both in the qualitative manner and quantitative manner. In Section VI, we discuss the relationship between our proposed method and previous studies and analyze the main limitations of current method. Finally, we conclude this article in Section VII.

II. MATERIALS

A. Data Sets

In our study, we totally collected 15 CEUS cine loops from patients (F/M = 5/10) who attended Nanjing Drum tower hospital for liver lesion examination from September 2018 to April 2019. The median patient age was 54 years (ranging from 37 to 73 years) and the proportion of body fat is 46.7%. Fifteen patients with hepatocellular carcinoma (HCC, 12 of 15) and intrahepatic cholangiocellular carcinoma (ICC, 3 of 15) were included. In B-mode tissue images, motion-related structures that can be observed include the liver diaphragm that is located directly beneath the liver, small vessels, part of high-contrast lesion boundary, and so on. Each examination was performed by an expert radiologist with over ten-year clinical manipulation experience. All examinations were performed on a Logiq E9 ultrasound scanner (GE Healthcare, Milwaukee, WI, USA) with a low mechanical index (MI) set below 0.15. The CA for CEUS was microbubbles filled with SonoVue (Bracco, Milan, Italy). A 1.2-mL bolus of CA was injected into the cubital vein, followed by the injection of 5-mL saline solution. Approval was obtained by the ethics review board of local hospital, and the informed consent was obtained from patients before this study.

Each CEUS cine loop is acquired in dual modes, i.e., the conventional BUS mode and the contrast-specific mode simultaneously displayed side by side, and both modes were recorded with the same duration and framerate. Each recording started right after the injection of contrast and lasted about 2 min at a frame rate of 8 frames/s. The spatial resolution of both views is 300×400 pixels. In order to ensure the consistency among different examinations, all selected CEUS sequences are acquired at the same microbubble concentration and mechanical parameters. Apart from that, all patients were allowed to breathe freely without holding breath intentionally, radiologists were required to hold the probe in a relatively fixed position, and the whole perfusion process was recorded at the same field of view (FOV).

Irregular Respiratory Motion: In our collected CEUS data, tissue displacements are mainly caused by free-breathing and cardiac movements. Both breathing depth and frequency are apt to be disturbed by possible physiologic variations [13], [29], and the maximal amplitude of diaphragm displacements was up to 30 mm. Furthermore, according to the visibility of dominant moving structures (e.g., liver diaphragm, vessels, and high-contrast lesion boundary) in the successive BUS images, we divided all sequences into two groups, Group1

(nine cases with clearly visible structures) and Group2 (six cases without such structures).

B. Data Preprocessing

To reduce the computational burden, we rescaled the CEUS (BUS) from the original resolution 300×400 to 150×200 without changing FOV and temporally down-sampled the original sequence by a factor 2. The resulting resolution is sufficient for our RCME method and the comparative approaches. In addition, to reduce the impact of speckle noise in B-mode images, we adopt an adaptive filter proposed in [31], which adapts the kernel of filter to fit homogeneous regions using local image statistics.

III. METHOD

In this section, we introduce in detail our proposed RCME method, including the architecture of our network, the manifold theory and optimization of OT, the clustering-based frame selection mechanism, and the implementation details.

A. Architecture

Like most respiratory compensation algorithms, our proposed RCME method is performed on the B-mode image space. As shown in Fig. 2, it consists of two main parts: 1) OT-based motion estimation and 2) frame selection via motion clustering.

Briefly, the input dual-mode CEUS cine loop is first split into grayscale BUS time series I_i^{bus} and the accompanying contrast-specific CEUS sequence I_i^{ceus} ($i = 1, 2, \dots, N$), $I_i^{\text{bus}}, I_i^{\text{ceus}} \in R^{n_x \times n_y}$, where n_x and n_y are the height and width of BUS (CEUS) frames, respectively, and N is the total number of successive frames. Then, OT projects the input successive BUS frames into manifold space and estimates the motion information with respect to the template, based on which, the subset S of BUS (CEUS) with the same 2-D plane displacements is identified to restore motion-free TIC for subsequent quantitative analysis. Each of these steps is detailed in the following context.

B. Transport-Based Motion Estimation

1) *OT and Manifold Projection:* Since the ability to jointly model pixel intensities as well as their locations [32], OT has been applied in the morphologic analysis of medical imaging, for example, brain morphology difference analysis [30] and visualization of malignant tumor cells [33]. The goal of OT is to find the optimal intensity matches, which morphs one tissue image to another at the minimal transport cost. From the perspective of geometry, the OT distance of tissue spatial distribution defines not only a metric space but also a Riemannian manifold that is consistent with the continuous irregular liver motion, and successive B-mode images can be considered to move along the geodesic curve of the manifold [32], [34].

Since the OT distance is calculated among probability measures, we first normalize original grayscale BUS frames I_i^{bus} ,

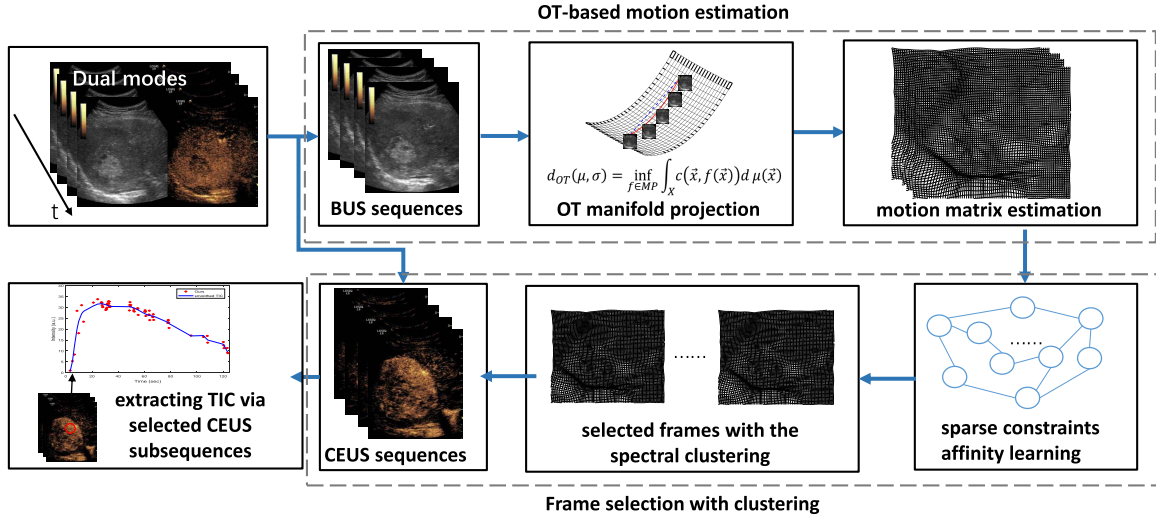


Fig. 2. Overview of our proposed RCME method for irregular respiratory motion compensation. To tackle the challenge of motion irregularities, our method first utilizes OT to estimate the salient motion information of dominant structures by jointly comparing pixel intensities as well as their locations. Then, SSC is employed to group successive frames based on the estimated motion matrix. Finally, motion-free TIC is reconstructed using the selected contrast-specific frames.

resulting in smooth density functions that compare them at full resolution, $\int_{\Omega} I_i(\vec{x}) d\vec{x} = 1$, where $\Omega = [0, 1]^2$. Considering two probability measures μ and σ associated with smooth density functions I_i^{bus} and I_j^{bus} , we have $d\mu(\vec{x}) = I_i^{\text{bus}}(\vec{x}) d\vec{x}$ and $d\sigma(\vec{x}) = I_j^{\text{bus}}(\vec{x}) d\vec{x}$. Here, μ and σ are defined on metric space X and Y (i.e., $X, Y \in \Omega$). The goal of OT problem is to find the optimal intensity mapping f that pushes μ on to σ at the minimal transport cost, which can be formulated as

$$d_{OT}(\mu, \sigma) = \inf_{f \in MP} \int_X c(\vec{x}, f(\vec{x})) d\mu(\vec{x}) \quad (1)$$

where $c : X \times Y \rightarrow R^+$ denotes the ground transport cost, and here, the L^2 norm of 2-D image coordinate is adopted, $c(\vec{x}, f(\vec{x})) = \|\vec{x} - f(\vec{x})\|_2^2$. MP denotes the set of mass-preserving mapping satisfying the following constraint $MP := \{f : \int_{f^{-1}(A)} d\mu = \int_A d\sigma\}$ for any measurable $A \subset Y$.

For smooth density functions, f is one-to-one mapping, which means that the mass at one position $\vec{x}$ can only be transported to a unique location $f(\vec{x})$. Besides, the mass-preserving constraint can also be written in differential form

$$\det(D_f(\vec{x})) I_j(f(\vec{x})) = I_i(\vec{x}) \quad (2)$$

where $\det(D_f(\cdot))$ represents the determinant of Jacobian matrix of f . It is worth noting that mass preservation constraint is applicable for the movement estimation of major anatomical structures (e.g., like the liver diaphragm) since deformation induced by respiration basically satisfies the mass-preservation assumption [35].

More importantly, OT distance d_{OT} together with the set of continuous probability measures $P(\Omega)$ geometrically constructs a Riemannian manifold $(P(\Omega), d_{OT})$, where any point defines a tangent space equipped with inner product [36]. As shown in Fig. 3, for any two moving BUS frames, there exists a geodesic path representing how one frame evolves into another [37], which explicitly models the successive 2-D respiratory motion in BUS time series.

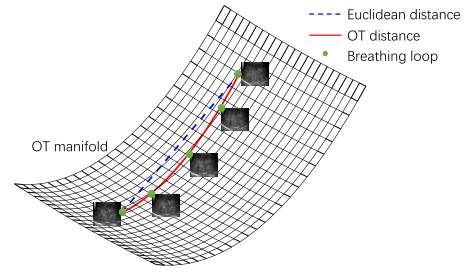


Fig. 3. Visualization of manifold structure of continuous BUS frames in a half breathing loop modeled by OT. Red solid line: geodesic curve that represents how inspiratory frame moves to expiratory one in OT distance space. Blue dashed line: moving path in Euclidean space. Small green circles: certain frames in the breathing loop.

2) Motion Matrix Estimation: Given a set of continuous BUS images $I_i^{\text{bus}} (i = 1, 2, \dots, N)$ and a template frame I_0^{bus} , we have the corresponding smooth densities μ_i and σ , satisfying $d\mu_i(\vec{x}) = I_i(\vec{x}) d\vec{x}$ and $d\sigma(\vec{x}) = I_0(\vec{x}) d\vec{x}$. The OT map between μ_i and σ can be obtained by the constrained minimization problem

$$f_i^*(\vec{x}) = \arg \min_{f_i} \int_{\Omega} \|f_i(\vec{x}) - \vec{x}\|^2 I_0(\vec{x}) d(\vec{x}) \quad \text{s.t. } \det(D_{f_i}(\vec{x})) I_i(f_i(\vec{x})) = I_0(\vec{x}). \quad (3)$$

Notably, optimal intensities matching in (3) are performed at each pixel, $u_i(\vec{x}) = f_i^*(\vec{x}) - \vec{x}$ denotes the 2-D displacement with respect to the template I_0^{bus} at pixel $\vec{x}$, and $[u_i^x(\vec{x}), u_i^y(\vec{x})]$ denote the x -axis and y -axis displacement, respectively.

In our study, we employ a flow minimization scheme AHT proposed in [38], to obtain the optimal mass preserving mapping f_i^* (i.e., mass intensities matching). This optimization strategy is built on the fundamental theoretical result [39], which states that the unique solution to (3) is curl free

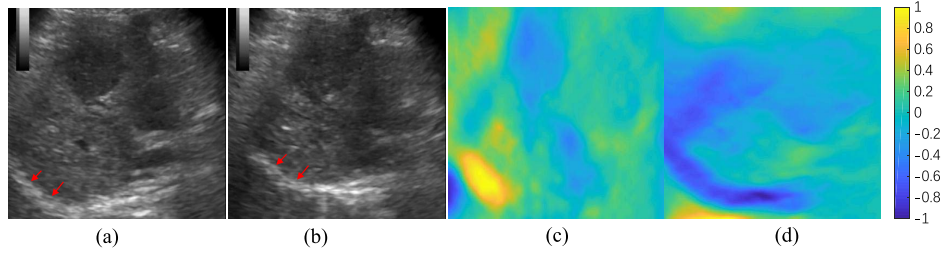


Fig. 4. Example of OT-based motion estimation. (a) Moving frame. (b) Reference frame. (c) Color map of the pixelwise displacement along the x -axis, u_i^x . (d) Pixel-level displacement along the y -axis, u_i^y . For the better visualization, we rescaled the estimated motion vectors u_i to range $[-1, 1]$. Red arrows: region containing obvious displacements.

as it can be written as the gradient of a convex function $f_i^* = \nabla p$. Specifically, this mechanism first obtains an initial mass preserving transport map f_0 using the Knothe–Rosenblatt coupling and then updates the initial mapping via gradient descent to make f_0 curl free [35]. AHT defines $s(x, t)$ for time step t , and satisfying $f := f_0(s^{-1}(\tilde{x}, t))$, $s(x, t_0)$ is a mapping from μ to itself. Starting from $s(\tilde{x}, 0) = \tilde{x}$, the initial transport map f_0 is updated to converge to a curl-free mapping f_i^* . Detail derivation can be found in [35].

From the perspective of feature extraction, grayscale tissue imaging is not represented by raw intensity but the salient motion information. Fig. 4 gives a visualization of quantified motion matrix, where major displacements can be clearly observed even when gray anatomical structures seem ambiguous. In addition, OT-derived transformation significantly enhances interframe separability compared with the original pixel domain [30], which is critical to the subsequent clustering-based subsequence extraction.

C. Frame Selection via SSC

With the given motion matrix sequence $M = \{M_i | M_i = (U_i^x, U_i^y), i = 1, 2, \dots, N\}$, where $U_i^x(U_i^y)$ is the 2-D matrix quantifying displacement of each pixel along the x -axis (y -axis) direction, the following step is to extract a subset of frames from the same reference position. In principle, any existing clustering algorithm could be used to acquire the target subset. Considering the strong structural similarity of frames at the adjacent positions, the high-dimensional motion matrix sequence essentially lies on multiple separate low-dimensional manifolds. In order to mine the multisubspace structure, we choose the SSC to group the motion sequence into different clusters.

For SSC, the affinity graph matrix W is constructed on the sparse self-representation (SSR) learning of M . W is expected to encode the structural similarity pattern of the periodic and irregular organ motion. SSR is well suitable to uncover the separate multisubspace structure, in that sparse constraint of representation coefficient c_i enforces each point to be a linear combination of a sparse set of D from the neighboring position. Specifically, we flatten the paired displacement matrix (U_i^x, U_i^y) and concatenate them into a single motion vector $m_i \in R^{2L}$, L is the number of pixels of detected high-contrast anatomical structures, dictionary D is the original irregular motion matrix sequence, and learning of sparse coefficients c_i

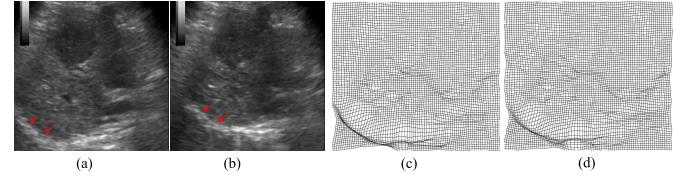


Fig. 5. Recovered motion information with sparse representation. (a) Moving frame. (b) Reference frame. (c) Grid of the original motion matrix. (d) Grid of the sparse recovery version. Red arrows point to dominant moving structures.

can be written as follows:

$$\begin{aligned} \min_{c_i} & \|Dc_i - m_i\|_F + \lambda \|c_i\|_1 \\ \text{s.t. } & c_i \geq 0, \quad c_{ii} = 0 \end{aligned} \quad (4)$$

where λ balances the representation residual error and sparse regularization. In our implementation, λ is determined through a grid search ranging from 0.1 to 0.3 at a step of 0.05. In this way, both the intrasubspace similarity and the intersubspace difference are expected to be enlarged, thus obtaining a reliable coefficient matrix C . Next, we normalize C with each column being $c'_i = c_i / \max(c_i)$ to construct the affinity graph matrix $W = |C'| + |C'|^T$. Fig. 5 shows an example of recovered motion information with sparse representation. Besides, for a better illustration of the improvement brought by OT, Fig. 6 gives a visualization comparison of the affinity graph matrix W . As can be seen, Fig. 6(b) with transport-based motion estimation preserves much more interframe similarity (top left) than that of original intensity space [shown in Fig. 6(a)], which can be explained by the manifold projection of OT-derived transformation [30]. This is critical for motion compensation because preserving contrast frames from the same reference position but distant breathing cycles can help us retain more perfusion details, especially for the interval around perfusion peak.

Then, the affinity matrix W is applied for the subsequent spectral clustering algorithm (normalized graph cuts). Three main steps are detailed as follows. First, construct the normalized graph Laplacian matrix L as $L = A^{-(1/2)}(A - W)A^{-(1/2)} = I - A^{-(1/2)}WA^{-(1/2)}$, where A is the diagonal matrix whose i_{th} element is the sum of i_{th} row of W . Second, compute the first k eigenvectors $U = [u_1, u_2, u_3, \dots, u_k] \in R^{N \times k}$ of L and normalize each column of U to have unit length. Third, each row of U is treated as a new k dimensional

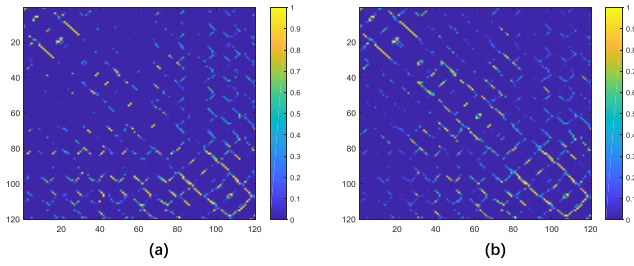


Fig. 6. Visualization comparison of the affinity graph matrix W . (a) Affinity graph matrix derived from original intensity space. (b) Affinity graph matrix estimated from the motion feature space via OT. As can be observed, more interframe similarity (top left) can be efficiently preserved to restore perfusion details.

vector representation of original motion matrix M_i and fed into k -means algorithm. For simplicity, k is the set equal to the cluster number K . Finally, we extract the target image cluster C_i corresponding to reference position, and the changing of contrast concentration over time is recovered based on the synchronized subsequence of the contrast-specific mode.

D. Implementation

1) *Selection of Reference Position*: Similar to respiratory gating methods, we need to specify a certain lesion position (i.e., reference position) for CEUS subsequence extraction. Generally, the reference position is chosen as extreme phases so that more contrast frames could be preserved to describe the intensity changing process due to much slower tissue movements [13]. In addition, considering irregular respiratory motion pattern, we selected the end-of-expiration point as the reference position in most breathing loops. Empirically, we have chosen the template frame from the end-of-inhalation phase (opposed to the selection of reference position) such that relatively larger pixel movements are insensitive to noisy displacements and beneficial to sparse representation learning.

2) *TIC Ground Truth*: For obtaining ground-truth TICs, we apply the template matching [40] for tracking the liver diaphragm that is located beneath the liver. Based on the tracking result, we select the subset of contrast frames corresponding to the same reference position specified as discussed above. Certainly, manual correction of false selection is necessary since diaphragm tracking itself cannot guarantee reliability. Tracking is implemented in Group1 for relative clear diaphragm structures. Besides, the ROIs were outlined manually according to the perfusion pattern at the arterial and late portal phase. Generally, ROIs can be placed within the lesion totally or partly overlap the surrounding normal tissues. It is noted that, due to the absence of a well-distinguished liver diaphragm in Group2, we could not perform the diaphragm tracking to acquire the ground-truth TICs. Thus, the performance of motion compensation in Group2 was evaluated by the smoothing indexes R^2 and $RMSE$ detailed as follows.

IV. EXPERIMENTAL SETTING

In this section, we first give a brief introduction of the used qualitative and quantitative evaluation methods, including the parametric perfusion imaging (PPI), the quality of curve

fitting, and the quantitative distance defined on fitting coefficients. Then, we describe different types of competing methods as well as their implementation details. Further, we design a group of ablation experiments to verify the effectiveness of this combination of OT and SSC for irregular liver motion compensation.

A. PPI

PPI, an efficient multiparametric functional imaging formed by pixelwise TIC kinetic parameters, is used to quantify hemodynamics in microvessels [41], [42]. In contrast to regional TIC analysis, PPI has the advantage of preserving spatial information of hemodynamic distribution in lesions or surrounding tissues. However, pixelwise kinetic parameters estimation is more vulnerable to cyclic liver motion and hemodynamics differences between the lesion and normal tissues could be greatly reduced. Therefore, PPIs formed with or without motion correction can be employed as a qualitative assessment tool for motion compensation strategy [43]. In our study, perfusion parameters include two intensity relevant parameters of PV and AUC and time-related indexes of mean transit time (MTT) [8].

B. Evaluation of Respiratory Compensation Performance

In particular, the qualitative and quantitative evaluation methods are described as follows.

- 1) *Qualitative Assessment With PPIs*: PPIs of multiple perfusion parameters, such as PV, MTT, and AUC, were formed with or without respiratory compensation, respectively. The effectiveness of our proposed method can be visually assessed by comparing how much perfusion detail is preserved around the area of lesion and surrounding tissue.
- 2) *Quantitative Evaluation of the Compensated TICs*: A common assumption is that ideal regional or pixelwise TICs without artifacts should be consistent with logarithmic function that models the CA wash-in and wash-out process. Therefore, the quality of TIC fitted to the logarithmic model can be taken as quantitative indexes.

In our experiments, we use the gamma-variate function as a fitting model [18]

$$c_{\text{gamma}}(t) = \begin{cases} b, & t < t_0 \\ b + a \cdot (t - t_0)^\alpha \cdot e^{-\frac{t-t_0}{\beta}}, & t \geq t_0 \end{cases} \quad (5)$$

where t_0 represents the contrast arrival time, α and β are two parameters that model the raise and wash-out of contrast, and a and b are determined by PI and background intensity, respectively.

The quality of TIC fitting is quantitatively measured by two indexes, the coefficient of determination R^2 and the root mean square error RMSE, which are defined as [18]

$$R^2 = 1 - \frac{\sum_i (y_i - c_i)^2}{\sum_i (y_i - \bar{y}_i)^2} \quad (6)$$

$$\text{RMSE} = \sqrt{\frac{\sum_{i=1}^n (y_i - c_i)^2}{n}} \quad (7)$$

where y_i are the intensity of the obtained TICs and c_i are the corresponding value from the fitted model. $\bar{y}_i$ is the mean value, and n is the number of time points.

Besides, to quantitatively compare the difference of corrected regional TIC and ground truth obtained by diaphragm tracking, we computed the Euclidean distance of two coefficient vectors $Ed = (1/4)\|f_{\text{corrected_TIC}} - f_{\text{GT_TIC}}\|_2$, and their gamma-variate functions are fitted by corrected TIC and ground-truth TIC. In our experiment, the accuracy of TIC was only evaluated at the regional level in which we consider that diaphragm tracking is not accurate enough to serve as ground truth at the pixel level.

C. Competing Methods

In this work, our proposed RCME method was compared with several mainstream respiratory compensation methods, including a registration method aligning anatomical structures across the whole CEUS sequence, i.e., block matching (BM) [23], a gating method estimating respiratory curves at patch level with multiple resolutions, i.e., multiresolution PCA [27], and an integrated method fusing NMF-based breathing curves estimation and BM [28].

1) *Registration-Based Method (BM)*: In our study, we implemented the BM algorithm proposed in [23], which adopts an adaptive rood pattern search (ARPS) strategy to accelerate tissues matching, which assumes that neighboring blocks share similar motion vectors. BM algorithm was performed with the similarity metric of mean absolute difference (MAD). The block size s is set to four pixels and the search area $p = 10$ pixels is centered at the block in the previous frame. To avoid the accumulation of matching errors, the interval of continuous correction I is set to 8.

2) *Respiratory Gating-Based Method (Multiresolution PCA)*: In line with [27], we first divided original BUS time series into multiple equal-size patches series, that are used to obtain respiratory curves using PCA, and then grouped them using hierarchical clustering. Finally, the breathing curve is selected as the median prototype from the largest cluster. Following most gating methods, the first three principal components are used to restore breathing signals. The minimal size of the largest cluster Δ_m is set to 8 and the sizes of partitioned patch $Q = \{8, 16, 24, 32\}$.

3) *Integrated Method Combining Breathing Gating and Registration (rMoCo)*: In the rMoCo, breathing curve was first estimated using NMF. After that, the obtained curve was divided equally to generate dozens of subsequences, and finally, block matching (APRS) is used to align the first image of each subsequence. Following the same experimental setting, the number of kinetic factors is set to 2 and the breathing signal is obtained by normalizing the first factor. The iteration process would not terminate until the minimal Euclidian difference 10^{-3} or the maximal iteration number 120 is reached. The number of selected subsequences (phases) is set 15. Parameters selection of BM follows the same steps aforementioned.

Notably, all these competing methods as well as our proposed RCME are implemented in the simultaneous B-mode sequence. Pixel- or regional-level TICs are reconstructed by fitting pixel intensities of the preserved contrast frames using

the lognormal-gamma model. As for PCA or NMF-based breathing gating methods, kinetic values higher or lower than the mean by more than 55% were considered as outliers caused by out-of-plane motion and should be removed before TICs fitting. Details on how to select optimal parameters for each competing method are given in Table S1 in the Supplementary Material.

D. Ablation Experiments

In order to validate the effectiveness of the important components of our method, we also performed another two groups of ablation experiments, which perform the OT or SSC only for irregular motion compensation. Besides, we also compared another type of motion estimation method, optical flow (OF) [44], to test the superiority of OT-based salient displacement estimation.

V. RESULTS

In this section, we first present the qualitative compensation results on cases of HCC and ICC. Then we present the quantitative comparisons with several competing compensation methods. After that, we further evaluate the influence of several experimental setting (e.g., the selection of template frame I_0^{bus} and the clustering number K). Notably, due to the absence of the corresponding ground truth TICs in Group2, only the smoothing metrics (i.e., R^2 and RMSE) can be used to evaluate compensation performance. Thus, we choose Group1 for these two groups of experiments such that all three quantitative metrics could be used for a more comprehensive analysis of different parameter settings. Finally, we verify the effectiveness of this combination of OT and SSC by an extra group of ablation experiments.

A. Qualitative Assessment With PPIs

As functional imaging, PPI formed by the pixelwise kinetic parameters is able to depict the distribution of microvascular hemodynamics without loss of spatial resolution. Fig. 7 shows several comparative PPI views before and after motion compensation using our proposed method. The presented PPIs are from two cases of HCC and one case of ICC, and the functional parameters are of particular importance to clinical diagnosis with respect to the lesion type.

By definition, MTT refers to the time interval when pixel intensity is above the half of the peak value, and thus, the respective view records the heterogeneous distribution of flow velocity of CAs. As observed from the first row in Fig. 7, most MTT values within the lesion (denoted by red contour) are around 35 s after respiratory correction, while the majority of MTT values without compensation are over 42 s. It is clear that our proposed motion compensation method enables to recover more details on inhomogeneous velocity distribution compared with that of uncompensated PPI. The main reason is that a large proportional of intensity variations within lesion is the mixture of intensities from adjacent tissues where the fade of CA is much slower. Another parametric map AUC encodes the perfusion volume at each pixel, which integrates the local concentration variation over the whole perfusion

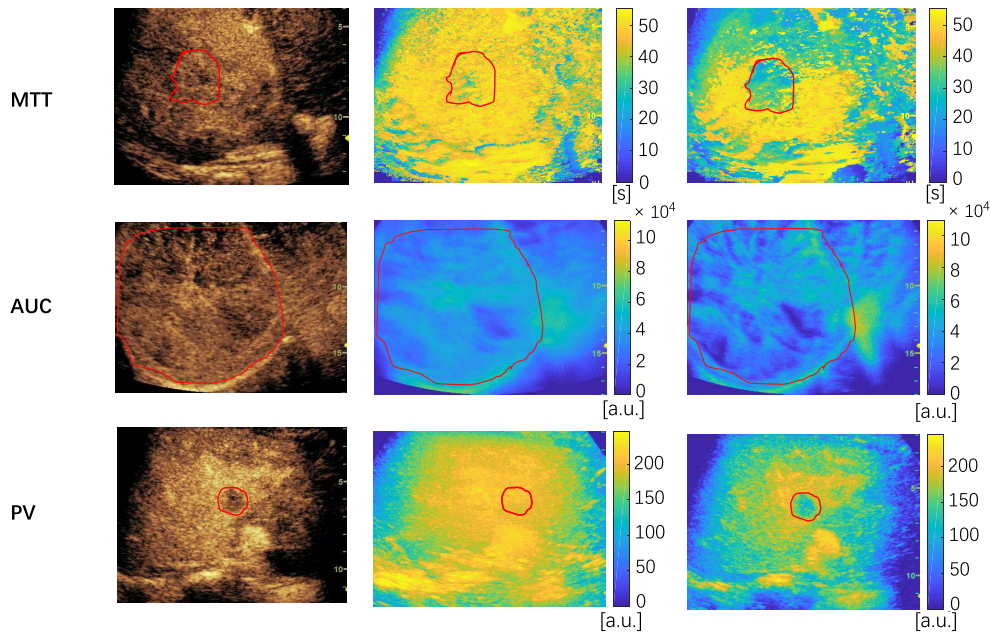


Fig. 7. Examples of comparison of PPIs formed with and without motion correction. The first column shows the representative CEUS frames, PPIs of the second column are formed without breathing motion correction, and the third column shows the compensated PPIs using our proposed method. The first two rows correspond to two examples of HCC case and the third row presents the ICC case.

process. Similarly, the compensated AUC map allows for clear observation of wheel-like vascular distribution. As known from the clinical guidance, hypoenhancement in arterial phase is one of the typical perfusion characteristics for ICC identification. In Fig. 7, the relatively small ICC lesion is nearly invisible in PV maps formed with original CEUS sequence, due to the repetitive and irregular liver movement. In contrast, the corrected map preserves the hypo-enhancement pattern of ICC lesion and the lesion boundary is easily to be delineated.

To sum up, irregular liver motion causes pixelwise TICs to contain intensity from pixels of neighboring tissues, leading to the loss of information on inhomogeneous hemodynamic distributions. Our proposed motion correction method is capable of recovering the missing information by eliminating the interference from uncorrelated tissues.

B. Quantitative Evaluation of the Compensated TICs

1) *Example of TIC Fitting*: Here, we take one CEUS sequence of HCC as an example, and Fig. 8 gives an intuitive comparative result of the corrected TICs obtained by different methods. As shown in Fig. 8, ROI is placed in the internal of lesion and the diaphragm is visible. As can be observed, all respiratory correction methods enable to restore the characteristic of “fast wash-in/wash-out” that is typical to HCC. Also, TTP and PI are the most error-prone kinetic parameters due to the mixture of intensities adjacent tissues. Multiresolution PCA and NMF-based rMoCo are still subject to disturbance induced by several irregular breathing cycles even with a carefully selected threshold. The main reason is that some expiratory phase points with relatively smaller magnitudes are still be preserved, but the corresponding frames have not yet reached a reference position in most cycles. This is also the main

drawback of respiratory signal estimation-based methods for the case of irregular hepatic motion correction. In contrast, the smoothed TIC estimated from our proposed method has higher consistency with ground-truth signal and less abrupt intensity fluctuations. This can be explained by motion feature estimation using OT, which allows for a more sensitive distinction of adjacent frames.

2) *Quantitative Results of Clinical Data*: In our experiments, clinical collected CEUS cine loops are divided into two groups according to the existence of high-contrast anatomical landmarks (e.g., diaphragm located beneath the liver and vessel). Group1 contains nine patients with relatively clear and complete landmarks, whereas the other six patients are classified as Group2. As mentioned earlier, TIC ground truth (GT) can only be acquired in Group1 by tracking the visible diaphragm structure, and the quantitative results compared with the GT TICs are listed in Table I. Besides, the quantitative evaluation of TIC fitting quality is performed on all patients and the results are presented in Table II (i.e., R^2) and Table III (i.e., RMSE).

Several important observations can be summarized from Table I.

- 1) The impact of liver motion can be alleviated to some extent using various respiratory compensation methods, including the competing methods and our RCME method. This shows that irregular liver motion information, such as 2-D displacements and breathing phases, can be obtained in synchronous B-mode tissue images with clearly visible diaphragm structures.
- 2) Methods relying on keyframes extraction (i.e., Multiresolution PCA, rMoCo, and RCME) yield better results than the registration method (i.e., BM), demonstrating

TABLE I
QUANTITATIVE COMPARISON RESULT OF GROUP1 CEUS SEQUENCES WITH TIC GROUND TRUTH VIA DIAPHRAGM TRACKING

Sequence		Ed				
		Original	BM[23]	Multi-Resolution PCA[27]	rMoCo[28]	RCME
Group I	1	13.396	9.651	8.621	8.542	7.441
	2	10.405	6.712	4.157	4.431	4.123
	3	16.703	10.621	7.422	7.244	6.501
	4	21.513	13.451	9.142	6.179	5.911
	5	19.124	10.245	8.423	7.962	5.124
	6	26.694	13.421	6.297	7.610	7.458
	7	12.456	7.259	6.763	7.872	6.175
	8	25.660	16.910	10.652	12.765	8.740
	9	11.426	6.752	4.676	5.672	4.682
mean±std.		17.486±6.124	10.558±3.500	7.350±2.112	7.586±2.337	6.239±1.480*

* indicates that our proposed method is significantly better than the other four comparative methods (paired t-test at 95% significance level).

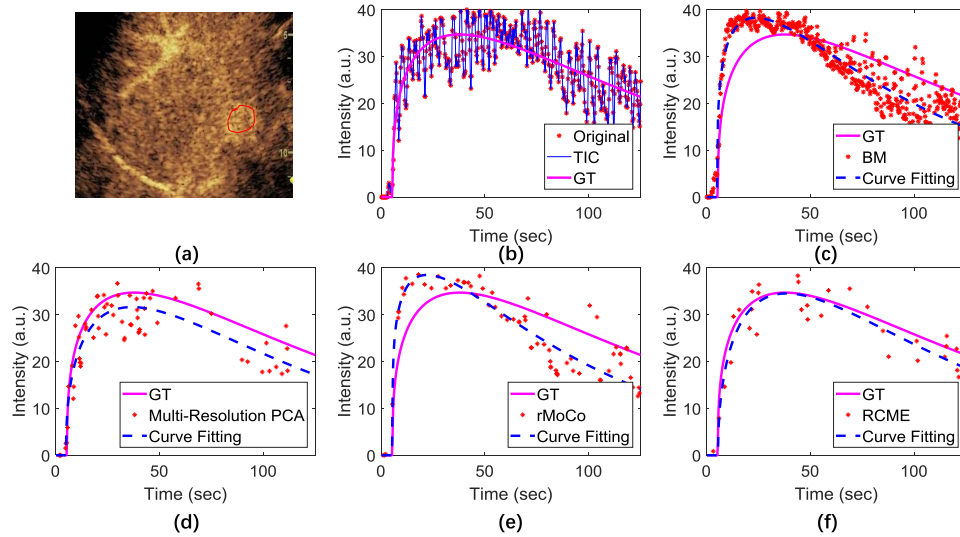


Fig. 8. Illustration of corrected TICs obtained by different respiration compensation methods. (a) Example of contrast frame (ROI denoted by red circle). (b) Original mean intensity (red dots) and the TIC (blue dotted line) generated from CEUS sequences. (c) Corrected mean intensity (red dots) and the fitted TIC (blue dotted line) using BM. (d) Corrected result obtained by multiresolution PCA. (e) Corrected result of rMoCo. (f) Corrected result of our proposed RCME method. Magenta line in each subplot denotes the regional TIC ground truth.

TABLE II
QUANTITATIVE COMPARISON OF TIC FITTING QUALITY R^2 OF TOTAL 15 CEUS SEQUENCES

Sequence		R^2				
		Original	BM[23]	Multi-Resolution PCA[27]	rMoCo[28]	RCME
Group1	1	0.672	0.716	0.828	0.847	0.925
	2	0.712	0.825	0.901	0.887	0.913
	3	0.620	0.696	0.756	0.766	0.869
	4	0.560	0.644	0.826	0.897	0.931
	5	0.601	0.670	0.781	0.816	0.911
	6	0.501	0.638	0.856	0.748	0.854
	7	0.697	0.724	0.829	0.799	0.922
	8	0.552	0.655	0.795	0.762	0.878
	9	0.703	0.744	0.935	0.850	0.945
mean±std.		0.624±0.076	0.701±0.060	0.834±0.057	0.819±0.055	0.905±0.031*
Group2	10	0.513	0.572	0.566	0.582	0.829
	11	0.627	0.652	0.674	0.633	0.897
	12	0.554	0.622	0.612	0.585	0.851
	13	0.455	0.534	0.551	0.594	0.787
	14	0.612	0.679	0.635	0.645	0.872
	15	0.422	0.566	0.575	0.551	0.795
mean±std.		0.531±0.083	0.621±0.045	0.602±0.047	0.598±0.035	0.839±0.043*

* indicates that our proposed method is significantly better than the other four comparative methods (paired t-test at 95% significance level).

that preserving critical points is more effective to restore original TICs with the help of perfusion curve fitting. Another reason could be that reliable deformation field estimation for aligning each frame is challenging, especially when considering the relatively

low ultrasound image contrast and artifacts interference around the liver lesion.

3) For most cases, our proposed clustering-based method RCME outperforms the methods (i.e., Multiresolution PCA and rMoCo) requiring explicit respiratory signal

TABLE III
QUANTITATIVE COMPARISON OF TIC FITTING QUALITY RMSE OF TOTAL 15 CEUS SEQUENCES

Sequence		RMSE				
		Original	BM[23]	Multi-Resolution PCA[27]	rMoCo[28]	RCME
Group1	1	4.812	4.160	3.941	3.281	2.063
	2	4.339	2.967	2.292	2.974	2.310
	3	6.403	5.373	4.648	4.522	2.416
	4	5.479	4.625	3.150	2.912	2.123
	5	5.435	4.982	3.884	3.147	1.872
	6	6.756	4.574	2.735	3.342	2.762
	7	5.045	4.018	2.683	3.061	2.110
	8	7.709	4.571	2.935	3.178	2.227
	9	5.053	4.716	2.165	2.757	2.097
mean±std.		5.670±1.754	4.443±0.683	3.159±0.833	3.208±0.542	2.220±0.256*
Group2	10	6.976	6.433	6.750	6.762	2.753
	11	5.483	5.299	4.952	5.396	2.201
	12	6.764	5.761	5.985	6.137	2.452
	13	9.112	6.855	6.825	5.810	3.072
	14	5.351	4.756	4.957	4.906	2.184
	15	9.608	6.147	5.885	6.845	2.885
mean±std.		7.216±1.792	5.850±0.774	5.892±0.821	5.976±0.763	2.591±0.369*

* indicates that our proposed method is significantly better than the other four comparative methods (paired t-test at 95% significance level).

estimation. This can be easily interpreted that the same breathing phase might not point to the same lesion position due to increased variability in respiration magnitude and frequency.

Therefore, the clustering-based method is more appropriate for addressing the challenge of respiration irregularity since most frames containing obvious deviation relative to the reference position are directly removed. The performance improvement brought by our proposed RCME is statistically significant (i.e., p -values < 0.05) since transport-based motion estimation allows for a much more effective similarity metric for moving frame clustering.

As observed from Tables II and III, for Group1, most methods can effectively enhance the TIC fitting quality both in terms of R^2 and RMSE by performing motion compensation. In contrast, the performance of multiresolution PCA and rMoCo deteriorated drastically for Group2, the average increase of R^2 for multiresolution PCA is only 0.071, and the average increase of R^2 for rMoCo is even lower 0.067, and the reduction of RMSE for multiresolution PCA is 1.324 and the reduction of RMS for rMoCo is 1.240, which is similar to the results of BM. It validates the importance of liver diaphragm as the guiding structure to generate the breathing signal, especially confronted with repetitive irregular liver motion. In Group2, although interframe difference degrades drastically on account of the ambiguity even invisibility of high-contrast structures, our proposed RCME still yields better fitting quality compared with the competing methods, the mean R^2 0.839 and the mean RMSE 2.591. It can be explained by the boosted interframe separability achieved by modeling the manifold structure of successive BUS sequence using OT.

C. Influence of Template Frame I_0^{bus}

To exploit the influence of I_0^{bus} on final TIC curves restoration, we add a group of experiments by selecting I_0^{bus} at the end-of-exhalation, middle position, and end-of-inhalation, respectively. This group of experiments is performed on Group1 (nine dynamic CEUS sequences included) with

TABLE IV
QUANTITATIVE RESULTS WITH DIFFERENT SELECTIONS OF TEMPLATE FRAME I_0^{bus}

Selection of I_0^{bus}	Ed	R^2	$RMSE$
End-of-exhalation	10.367±3.481	0.749±0.109	4.833±0.853
Middle	6.320±2.339	0.901±0.035	2.317±0.377
End-of-inhalation	6.109±2.631	0.924±0.047	2.149±0.301

clustering number $K = 16$. Quantitative results are shown in Table IV. Besides, to gain an intuitive understanding, Fig. 9 shows a qualitative comparison between the restored curves and the ground truth.

As can be observed, the best recovery result is achieved when I_0^{bus} is set to the end-of-inhalation (opposed to the setting of reference position), as shown in Fig. 9 and Table IV. Meanwhile, similar fitting performance could also be obtained at the middle position relative to the reference point. On the contrary, the accuracy of TIC recovery decreases dramatically when I_0^{bus} is set near the reference point, and considerable noisy points are preserved in the selected cluster. This can be explained that if I_0^{bus} is selected near the reference position, estimated displacements around the reference point would be much smaller, resulting motion vectors (near to 0) are prone to be represented by an all-zero coefficient under the sparsity constraint, leading to a biased similarity measure. Experimental results indicate that the final TIC recovery performance is not sensitive to the selection of template frame except for the neighborhood of reference point.

D. Influence of Cluster Number K

The clustering number K plays a key role in our RCME method, which directly determines the size of selected CEUS frames, N_{sel} . Apparently, N_{sel} would decrease dramatically as K is set to larger values. Although there is no need to preserve excessive contrast frames considering the redundancy of the enhancement process, too few frames might lead to the

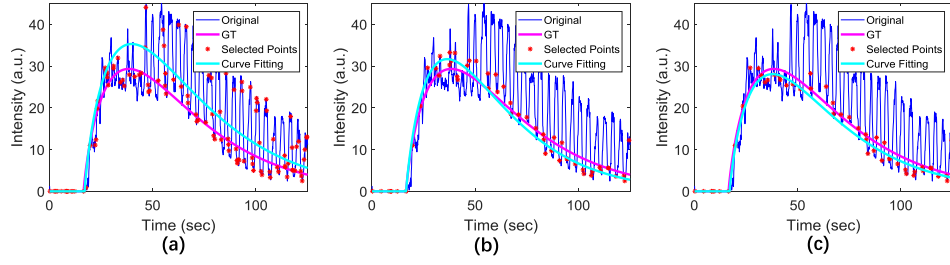


Fig. 9. Example of qualitative comparison of recovered TIC. (a)–(c) Recovered results when t_0^{bus} is set at the end-of-exhalation, middle position, and end-of-inhalation, respectively.

loss of valuable perfusion details. Hence, we added a group of experiments to exploit the influence of K . This group of experiment is performed on the Group1 sequence. Specifically, we tried the cluster number K from 2 to 24 and show the qualitative comparison with ground truth when K is set to 10, 13, and 16.

Fig. 10(a)–(d) shows the qualitative comparison results with ground truth of one subject. As can be observed, the basic variation tendency of contrast concentration can be preserved for $K = 10, 13, 16$. The best-fitting performance is achieved by setting K to 16 even only approximately 9% contrast frames are preserved. In contrast, for $K = 10$ and $K = 13$, TTP and peak value (PV) are easier to be biased compared with other parameters (e.g., AUC and MTT). The main reason is that more contrast disturbing frames around the neighborhood of reference points are maintained when contrast concentration reaches the peak.

Besides, Fig. 10(e) and (f) shows the influence of K on the mean number of preserved frames Δ and the degree of matching with ground truth Ed . The experimental result indicates that [14 – 18] is the appropriate interval to set clustering number K , and about 10% contrast frames are enough to restore TICs accurately. Accordingly, we propose a simple strategy for the selection of cluster number K in real applications, concretely, and the cluster number K is initialized as 18, decreasing by a step of 2 until the number of preserved frames Δ exceeds 10% of the original CEUS sequence.

E. Ablation Experiments

To validate the effectiveness of this combination of OT and SSC for irregular motion correction, in this group of experiments, we first test the compensation performance by performing the OT or SSC only, concretely, we replace the original SSC with baseline K -means. Besides, OT is also replaced by the classical OF [44] for the comparison of motion estimation method. This group of experiment is performed on the Group1 sequence with K setting to 16.

As shown in Table V, the highest recovery performance is obtained by our proposed RCME method, which validates the effectiveness of the combination of OT and SSC for irregular liver motion compensation. The combination of OF and SSC also yields better results than performing OT or SSC only, and this reflects the importance of motion estimation and multiple-subspace assumption to tackle the challenge of motion irregularity. When motion sequence clustering is only

TABLE V

QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS PERFORMED ON GROUP1 CEUS SEQUENCES

Method	Ed	R^2	$RMSE$
SSC	7.064 ± 2.029	0.836 ± 0.079	2.708 ± 0.579
OT + K-means	6.820 ± 2.339	0.841 ± 0.105	2.670 ± 0.977
Optical Flow + SSC	6.720 ± 1.639	0.867 ± 0.033	2.497 ± 0.271
RCME	6.239 ± 1.480	0.905 ± 0.031	2.220 ± 0.256

performed with K -means without multisubspace assumption, the final fitting results are only superior to performing SSC only. This indicates that the assumption of multisubspace structure is in accordance with the true distribution of motion sequence. In contrast, the worst performance is obtained by implementing SSC on original ultrasound images directly, and this can be explained by the greatly reduced interframe separability without any motion information.

VI. DISCUSSION

A. Comparison With Previous Work

Compared with the existing breathing gating methods, our proposed RCME method is specially designed to overcome the challenge of irregular liver motion by transport-based motion estimation. RCME takes the successive B-mode images as inputs to identify the subsequence acquired at the same position. In order to effectively filter those moving frames relative to the selected reference position, RCME employs the OT framework to estimate the 2-D displacements of landmark structures. In addition, frame selection is performed using SSC, in which sparse representation learning is used to suppress the disturbance (i.e., intensities mismatching) arising from artifacts or noninformative background tissue. In this way, we could avoid the main limitation of gating methods faced with the varying breathing pattern, namely, the same breathing phase cannot assure the same lesion position.

Our proposed compensation method is also different from those registration methods. Considering the known continuity of concentration variation of CA (e.g., the wash-in and wash-out stages), there is no requirement for aligning all frames across the whole CEUS sequence, and such group registration implementation inevitably increases the computational complexity, reducing the feasibility of

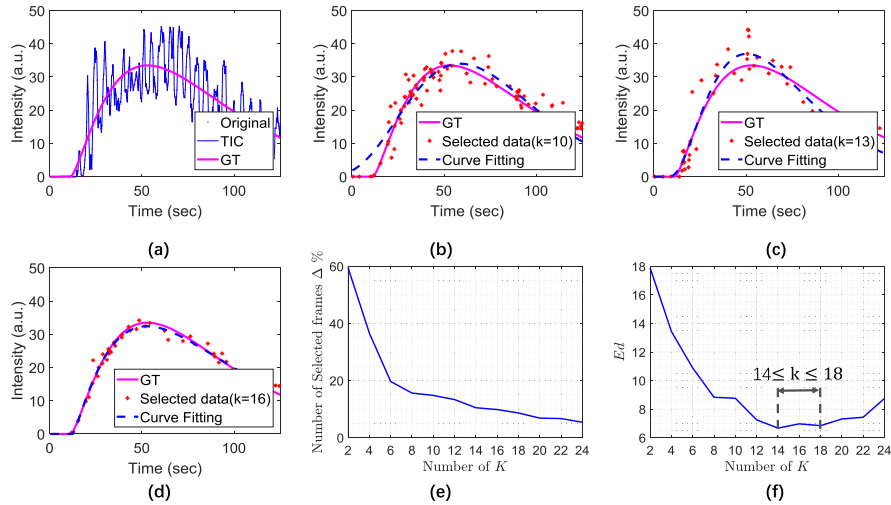


Fig. 10. Illustration of the influence of clustering number K . (a) Original TIC (blue dotted line) generated from uncompensated CEUS sequence. (b–d) Qualitative comparison when K is set to 10, 13, 16, respectively. (e) Influence of K on the mean number of preserved frames $\Delta\%$. (f) Influence of K on the mean degree of matching with ground truth E_d .

clinical application. Besides, accurate 2-D deformation field estimation might entail many difficulties, including the absence of landmarks caused by out-of-plane motion, disturbance resulting from artifacts or noises. It is worth to note that several outlier frames corresponding to out-of-plane motion would be removed in our RCME method because of the lower similarity. Moreover, the final curve fitting based on the logarithmic model can further eliminate the influence of possible wrongly selected frames.

Our experiments compared the performances of different respiratory compensation approaches for CEUS perfusion analysis. Overall, frame selection methods (gating methods and our proposed method) are superior to BM for most cases. For irregular respiration, the performances of motion estimation for Group2 deteriorates drastically compared with that of Group1, which indicates that the visibility of high-contrast landmark structures (like diaphragm) is of importance to various motion compensation methods. This observation inspires that during the clinical CEUS examination, ultrasound probe should be placed in a proper position where high-contrast anatomical landmarks could be kept visible in the 2-D imaging plane.

Besides, we also explored the sensitivity of the cluster size K of our proposed method. As the experiment indicates that, the number of selected frames decreases as the cluster size is set to larger values. If K is set too small, restored TIC would inevitably contain plenty of undesirable points in which interframe variance becomes too big within a cluster. In contrast, a larger K setting means fewer frames selected, leading to the loss of some perfusion detail. Nonetheless, for most cases, as long as the basic TIC shape is preserved, estimation of perfusion parameters is not much different. Ordinarily, there are approximately 10%–12% frames selected for reconstructing TICs. Therefore, the exact K value search

is not critical for TIC quantitative analysis, which increases the applicability for practical CEUS analysis.

B. Limitations and Future Work

While our proposed RCME method can achieve better results in irregular liver motion compensation, its performance could be further improved in the future by carefully dealing with some details in motion estimation. As known from our experiments, motion information of those high-contrast landmark structures plays an important role in interframe similarity measure as well as subset selection, while displacement estimation of some tissues that might not meet the constraint of mass preservation tends to be ineffective even disruptive. Thus, it is reasonable to highlight salient motion information by restricting OT mapping within salient tissues (e.g., liver diaphragm, small vessel, and so on). Apart from reducing computational cost considerably, it also helps reduce the amount of noisy displacements from speckle noises in the background area. To further eliminate the possible intensities mismatching caused by disturbing tissues, we wish to design a more effective disturbance removal mechanism before motion clustering. As for clinical application, there are still some limitations of our method. First, we might meet the case that rapid shadow breathing occurs in successive breathing cycles such that a few contrast frames could be preserved during this interval. This is mainly because our template frame is selected from the end-of-inhalation in most cycles. If this interval happens to be around the perfusion peak, the restored TICs would lead to a biased estimation of PI as well as the rate of contrast wash-in. Second, the construction of PPI depends on pixelwise TIC restoration, which inevitably results in much heavy computational load for clinical resources. When radiologists care more about the perfusion details within lesions, faster TIC restoration should be considered. One possible

solution is to take the correlation of neighboring tissues full into account.

VII. CONCLUSION

In this study, a motion estimation-based respiration compensation method was proposed to correct irregular liver motion for CEUS quantitative analysis. For alleviating the impact of motion irregularities, our method first introduces the OT to perform motion estimation by modeling the manifold structure of B-mode tissue sequences, based on which those spatially deviated frames are discarded directly by motion clustering, which is more applicable to overcome the challenge arising from the varying magnitude and frequency of liver movement. Our method is implemented in the synchronous B-mode tissue imaging, and the comparative results in two groups indicate the importance of high-contrast landmarks in the task of motion compensation for our method as well as the registration or gating-based approaches.

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Peng Wan received the M.S. degree in electronics and communication engineering from the Beijing University of Posts and Telecommunications, Beijing, China, in 2018. He is currently pursuing the Ph.D. degree with the College of Computer Science and Technology, Nanjing University of Aeronautics and Astronautics, Nanjing, China. His research interest is ultrasound imaging intelligent analysis, optimal transport, and machine learning.



Fang Chen is currently an Associate Professor with the Department of Computer Science and Engineering, Nanjing University of Aeronautics and Astronautics, Nanjing, China. Her research interests include medical image processing, image-guided surgery, and minimally invasive precision diagnosis and therapy.



Wei Shao received the Ph.D. degree from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2018. His current research interests include machine learning and bioinformatics.



Chunrui Liu received the M.S. degree in imaging and nuclear medicine from Nanjing University, Nanjing, China, in 2017. She has been a Sonographer with the Department of Ultrasound, Drum Tower Hospital, The Affiliated Hospital of Nanjing University Medical School, Nanjing, since 2017.



Yidan Zhang received the M.S. degree in imaging and nuclear medicine from Nanjing University, Nanjing, China, in 2018. She has been a Resident Doctor with the Department of Ultrasound, Nanjing Drum Tower Hospital, The Affiliated Hospital of Nanjing University Medical School, Nanjing, since 2018.



Baojie Wen received the M.S. degree in imaging and nuclear medicine from Liaoning Medical College, Jinzhou, China, in 2013. He has been an Attending Physician with the Department of Ultrasound, Nanjing Drum Tower Hospital, The Affiliated Hospital of Nanjing University Medical School, Nanjing, since 2013.



Wentao Kong received the M.S. degree in surgery from Nanjing University, Nanjing, China, in 2007, and the Ph.D. degree in imaging and nuclear medicine from Fudan University, Shanghai, China, in 2015. He has been an Associate Senior Doctor at the Department of Ultrasound, Drum Tower Hospital, The Affiliated Hospital of Nanjing University Medical School, Nanjing, since 2012. His professional emphasis includes contrast-enhanced ultrasound and interventional ultrasonography.



Daoqiang Zhang received the B.S. and Ph.D. degrees in computer science from the Nanjing University of Aeronautics and Astronautics (NUAA), Nanjing, China, in 1999 and 2004, respectively. In 2004, he joined the Department of Computer Science and Engineering, NUAA, as a Lecturer, where he is currently a Professor. His research interests include machine learning, pattern recognition, data mining, and medical image analysis. In these areas, he has published over 100 scientific articles in refereed international journals, such as the IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE, IEEE TRANSACTIONS ON MEDICAL IMAGING, IEEE TRANSACTIONS ON IMAGE PROCESSING, *NeuroImage*, *Medical Image Analysis*, *Pattern Recognition*, *Artificial Intelligence in Medicine*, and IEEE TRANSACTIONS ON NEURAL NETWORKS; and conference proceedings, such as the Conference on Neural Information Processing Systems, International Joint Conference on Artificial Intelligence, Association for the Advancement of Artificial Intelligence, SIAM International Conference on Data Mining (SDM), and International Conference on Data Mining. Dr. Zhang is a member of the Machine Learning Society of the Chinese Association of Artificial Intelligence (CAAI) and the Artificial Intelligence and Pattern Recognition Society of the China Computer Federation (CCF). He is a fellow of the International Association for Pattern Recognition (IAPR).